

The Story of FreeTraining-OVSS

How a project about recovering thrown-away pixels became a paper about the threshold that throws them away.

Full narrative — from the first commit to the last

Period — 11 August 2026 to 3 September 2026 (24 days)

Commits — 150

Code — 37 Python scripts, 7 shell scripts, 9,284 lines

Written record — ~60,000 words of measured results

Paper draft — 7,671 words, 26 references

Baselines reproduced — 3 of 3, all within 0.03 mIoU of published

Datasets — LoveDA, OpenEarthMap, ISPRS Potsdam

Hardware — one RTX 2000 Ada, 16 GB, power-capped at 70 W

The whole thing in one page

If you read nothing else, read this.

What we set out to do. Modern segmentation models are asked to label every pixel of a satellite image with a class name — *building, water, forest*. The best training-free method for this, SegEarth-OV3, has a rule: if the model is not confident enough about a pixel, throw that pixel into a bin called "background". We measured that bin. **On LoveDA it holds 29.68% of all the real land-cover pixels in the dataset** — 323 million of them. Our plan was to rescue them, using the fact that land cover has strong spatial grammar: roads touch buildings, buildings essentially never touch water.

What happened. The rescue plan failed, and it failed for a reason we could measure precisely. But while proving it failed, we found something better and simpler that works: **the confidence threshold itself is the wrong shape**. One threshold is used for all classes, and it is wrong for different classes in *opposite directions*. Fitting one threshold per class — using the same labelled data the baseline already spends tuning its single threshold, training no new weights — recovers a real gain.

THE FINAL CLAIM	SAM 3 PIPELINE (SEGEARTH-OV3)	CLIP PIPELINE (CONINFER)
LoveDA, 5-fold Δ mIoU	+1.18 $\pm$ 0.45	+2.51 $\pm$ 0.34
— worst fold of five	+0.80	+2.22
— excluding the catch-all class	+1.36	+1.94
Labelled tiles needed to calibrate	~200	~25
New model weights trained	none	none

WHY THIS MATTERS MORE THAN A BIGGER NUMBER WOULD

The gain is **larger on the pipeline it was not developed for**, and on that pipeline *every single class improves*. So the finding is not "SAM 3 has a quirk". It is a statement about **any** method that thresholds per-class confidence scores — which is nearly all of them. And it **composes with** our nearest published competitor instead of beating it: their score goes 36.99 $\rightarrow$ 39.52 when we put our calibration on top. We improve their method rather than compete with it.

The second half of the paper is a measurement study explaining *why* the residual exists and why rescuing it does not work. The short answer is that it is governed by **how the dataset was labelled**, not by the model. When a dataset provides a "catch-all" class that means *everything else*, the model always has a plausible answer available, and the discarded pixels become impossible to distinguish from genuine background. We proved that causally with a controlled experiment, and then a third dataset falsified half of it — which we report as a falsification rather than repair.

THE HONEST LEDGER

THING WE TRIED	OUTCOME
Co-occurrence prior (the original thesis)	✗ Dead. Worth +0.2 points; a <i>perfect</i> matrix is worth +0.3.
Just lower the threshold	✗ Costs 5.54 mIoU. 1.73 wrong pixels per 1 right.
Remove the presence gate	✗ Costs 11.97 mIoU. No net recovery at all.
Detect which pixels are recoverable	✗ Nine signals, all at chance on LoveDA.
Find a label-free rule for the thresholds	✗ Eleven attempts, all bounded, with the reason proved.
Better region shapes (SLIC superpixels)	✓ Ceiling 72.8% → 92.8%. Worth 20× the prior.
Per-class calibrated thresholds	✓ The method. Works on two architectures.
Catch-all-excluded mIoU as a reported metric	✓ Turned a "flat" dataset into a positive one.

Five of our own findings were refuted *by us*, before anyone else could. That is the part of the work I would point at first.

Act I — The wrong PDF

11-12 AUGUST · COMMITS `DA2BB6F` → `3CB1EDB`

The project began with a problem statement and two PDFs. The statement proposed an *unsupervised* pipeline: run a segmentation model, keep the confident pixels, build a matrix of which classes sit next to which, and use that matrix to label the unconfident pixels. Sensible. The repository at that point held a scaffold built on **SAM 1 + CLIP**.

Two things were wrong before a single line was run, and finding them in week one instead of week nine is most of why the project survived.

1. The second PDF was not the SAM paper

The file named `sam_research_papaer(1).pdf` was actually **SegEarth-OV3** (arXiv:2512.08730) — written by the same group that wrote the first PDF (SegEarth-OV, CVPR 2025). It already does a large part of what the problem statement proposed, using SAM 3 rather than SAM 1.

THE REALISATION

We had accidentally been handed **our own closest competitor**, and had been planning to rebuild half of it from scratch on an older model.

This was good news, not bad, and the reasoning matters:

- The baseline is **published and open source**. We do not have to build from zero.
- The question "does SAM 3 even work on satellite images?" is **already answered**. That saves a month.
- We now know **exactly** what our contribution has to be in order not to be a duplicate.

So the project was repositioned on day two. Not "an unsupervised SAM 3 pipeline", but: *SegEarth-OV3 throws away low-confidence pixels into background; we recover them*. SegEarth-OV3 became our **named baseline**, and our headline number became a delta against it rather than an unanchored score.

2. "Unsupervised" was the wrong word

The input includes a list of the possible class names. That is not unsupervised — it is **zero-shot / open-vocabulary / training-free**. SegEarth-OV itself footnotes that even "training-free" is a stretch for them, and settles on *annotation-free*. We adopted the same careful phrasing: **training-free, annotation-free open-vocabulary semantic segmentation**. Examiners and reviewers notice this.

Then: does the idea have any signal at all?

Before writing any method, we wrote a 40-line script (`cooccurrence_gt.py`) that measured, from ground-truth masks alone, whether land-cover classes actually have adjacency structure. The measure is **pointwise mutual information** (PMI): how much more often do two classes touch than they would if they were scattered at random?

MEASUREMENT	RESULT
Mean PMI over class pairs, LoveDA urban	1.658 bits
Mean PMI over class pairs, LoveDA rural	1.538 bits
Random-scatter control (what "no signal" looks like)	0.004 bits

Hundreds of times above the noise floor. The premise held. Three design decisions came straight out of that first measurement, and all three overturned what the original statement proposed:

ORIGINAL DESIGN	REPLACED BY, AND WHY
Adjacency = distance between region centres	Shared boundary length — two regions can be close without touching.
Add up positive evidence from neighbours	Signed PMI — the <i>largest</i> signals are negative. Building–water is $0.02\times$ chance. "This borders a building, so it is almost certainly not water" is the single most reliable fact available, and a positive-only formula cannot express it.
One matrix per image	Hierarchical mix of a corpus-wide matrix and a per-image one — 6 of 15 class pairs <i>flip sign</i> between urban and rural. A single global matrix averages them into mush.

And a second competitor appeared

ConInfer (arXiv:2603.29271, March 2026) claims to be the first work to use context at inference time for this exact task, reporting **+2.80 mIoU** over SegEarth-OV. Any reviewer in the area would know it. But it did not close our gap, and we wrote down precisely why:

	CONINFER	OURS (AS PLANNED THEN)
Backbone	CLIP	SAM 3
Kind of context	Visual (image features)	Semantic (class-pair adjacency)
Granularity	Patch level	Region level
Where the prior comes from	Fitted per scene	Mined across the corpus
Can express exclusion?	No	Yes (signed PMI)

Their own limitations section names pixel- and region-level context modelling as future work. That was our opening.

Act II — The week the environment nearly won

17–18 AUGUST · COMMITS `F18AD72` → `9EEA352`

Nothing in this act is intellectually interesting and it cost the most calendar time of any single obstacle. It is included because it is the honest answer to "why did week two take a week".

The three-way version deadlock

SegEarth-OV3 is built on MMSegmentation, which needs a compiled library called `mmcv`. Three requirements intersect at exactly one point in the whole version space:

COMPONENT	CONSTRAINT
SAM 3	needs torch $\geq$ 2.3 (it uses an API that did not exist before)
mmcv prebuilt wheels	exist for torch 2.1–2.4, and none for 2.5+
mmsegmentation 1.2.2	asserts mmcv < 2.2.0

So torch 2.4.1 is the *only* version that satisfies SAM 3 and still has a prebuilt mmcv. Its wheel is mmcv 2.2.0 — excluded by MMSegmentation's upper bound by one patch version. The fix was a one-character edit to an installed package (`MMCV_MAX = '2.2.0' → '2.3.0'`).

Five other approaches were tried and all failed:

- Python 3.13 — no mmcv wheels exist for it at all.
- `mmcv-lite` — MMSegmentation eagerly imports compiled CUDA operations at package load, so the lite build cannot work. Installing both breaks imports entirely.
- Building mmcv from source — the system CUDA compiler is version 13.3 while torch is built against 12.1, and torch refuses to compile across that gap.
- Installing the CUDA toolkit into the environment — the package that provides the compiler is a different one, so this quietly does nothing.
- Newer mmcv from source — its installer imports a module that modern setuptools removed.

STANDING RULE SINCE

Never run `pip install -U` in this environment, and never rebuild it by hand. The working combination is captured in `scripts/setup_env.sh`, and the project's instruction file states it in bold at the top. Every number in the project rests on it.

The baseline reproduced

Then the gate that made everything afterwards credible.

DATASET	PUBLISHED	OURS	DIFFERENCE
LoveDA (1,669 validation tiles)	47.4	47.38	0.02
OpenEarthMap (384 of 500 tiles)	42.9	44.19	+1.29
ISPRS Potsdam (2,016 tiles)	57.8	57.83	0.03

The OpenEarthMap row is on 77% of the official split, so it is a sanity anchor rather than a strict reproduction. The other two are exact.

Timing: 0.85 seconds per image, about 24 minutes for a full LoveDA pass, 6,115 MB peak memory of 16 GB available. The card runs at 100% utilisation against a **70 W power cap**, not a thermal limit — so 24 minutes is the floor on this hardware and no tuning improves it. That single fact shaped every later experimental design: we cached everything, so that later work became arithmetic instead of GPU runs.

The baseline's own table already showed the premise

Precision means "when the model says *water*, how often is it right". Recall means "of all the water that exists, how much did it find".

CLASS	IOU	PRECISION	RECALL	GAP
building	63.81	77.2	78.6	−1.4
road	53.89	69.6	70.5	−1.0
water	51.44	89.5	54.7	+34.8
agricultural	47.47	66.9	62.0	+4.9
background	45.50	56.9	69.4	−12.5
barren	35.73	51.5	53.9	−2.4
forest	33.78	57.9	44.8	+13.1

Read the two highlighted rows together. **Water: right nine times in ten when it fires, but finds only half the water present.** Those missing pixels are not being misclassified as something else — they are falling below the threshold and being dumped into background. And **background shows exactly the inverse pattern:** over-predicted and impure, absorbing pixels that belong to real classes. The bin is visible in the metrics.

Meanwhile *building* and *road* are balanced. The weakness is confined to amorphous "stuff" rather than sharp-boundary "things" — which is the exact duality SegEarth-OV3's own paper identifies. That told us where any gain would have to come from, seventeen days before we found the method that delivered it.

Act III — Measuring the motivation, and refuting ourselves twice

20-21 AUGUST · COMMITS [DF78A97](#) → [21F749E](#)

The premise needed a number, not an argument. We wrote instrumentation that records, for every tile, what fraction of real land-cover pixels the baseline assigns to background.

THE HEADLINE MOTIVATION NUMBER

29.68% of all real-class pixels in LoveDA validation are assigned to background — 323,184,908 pixels. The pre-agreed rule was that anything below 5% would mean pivoting the project. It cleared that bar by a factor of six.

The obvious objection, answered before a reviewer could raise it

Anyone reading that number asks the same thing immediately: *why not just lower the threshold?* So we ran it, at three settings.

THRESHOLD T	MIOU	REAL PIXELS DISCARDED
0.5 (the published setting)	47.37	29.68%
0.3	46.64	20.55%
0.1	41.83	10.88%

Lowering the threshold recovers two-thirds of the discarded pixels and costs **5.54 mIoU**. Tracking a single class through the sweep shows why: going from 0.5 to 0.1 recovers 85 million agricultural pixels and creates **148 million new false positives** — **1.73 wrong for every 1 right**.

And the cost is sharply non-linear, which is a subtlety we would have missed with a two-point sweep:

SEGMENT	PIXELS RECOVERED	MIOU COST	COST PER 100M PX
0.5 → 0.3	99.4M	-0.73	0.73
0.3 → 0.1	105.3M	-4.81	4.57

Nearly identical pixel counts; the second half costs **6.2 times more**. There is a shallow band just under the threshold that is cheap to reach, and a deep tail that is not. Any claim of the form "threshold relaxation is expensive" has to say *which part of the range*, or a reviewer with the middle row will object.

The dominant error is silence, not error

Every class's pixels end up in one of three places: correct, dumped in background, or confused with a different real class. Aggregated:

3 : 1

About **29.7%** of real-class pixels go to background against **9.9%** confused with another real class. **The baseline is mostly not wrong — it is silent**. That single ratio is the answer to "why not just improve the classifier", and it leads the results section of the paper.

Self-refutation #1: the presence head

SAM 3 has a "presence head" that outputs one number per class per image: *does this concept exist here at all?* SegEarth-OV3 multiplies every pixel score by it. We probed a tile that reported 100% discard, and found something that looked spectacular:

CLASS ON TILE 3487	PRESENCE SCORE	DENSE EVIDENCE	BEST POSSIBLE SCORE
road	0.0757	+10.13 (near-certain)	0.076
forest	0.0094	+5.03	0.009
agricultural	0.0200	+5.44	0.020

The dense part of the model is *certain* road is present. One global scalar crushes the whole tile below any usable threshold. It generalised, too: across all 1,669 tiles, catastrophic tiles had median presence **0.273** against **0.918** for healthy ones, correlation **−0.750**.

This looked like the strongest result in the project. We wrote down, *in advance*, what would count as proof and what would count as refutation, then ran the counterfactual: turn the presence gate off entirely.

TILE GROUP	DISCARD BEFORE	AFTER	CHANGE
catastrophic (198 tiles)	99.97%	60.82%	−39.15
healthy (77 tiles)	0.46%	54.11%	+53.64

REFUTED — BY US

Overall mIoU fell **11.97**. And the disqualifying number was not that one: the correlation between a tile's presence score and how much it recovered was **+0.018**. If gating were suppressing recoverable evidence, the *lowest*-presence tiles would recover *most*. It is zero. **The tiles that recover are not the presence-suppressed ones.** Presence collapse is a *correlate*, not a cause. No causal claim about it went into the paper.

The mechanism turned out to be something else entirely, and better. **SAM 3 essentially never detects "background"** — median presence **0.022**, against 0.45–0.91 for every real class. Obvious in hindsight: background is not a visual concept, it is the dataset's catch-all for unlabelled area. So gating was holding *background* down. Remove it and background surges and wins outright.

This is worth a sentence in the paper on its own, because it **inverts SegEarth-OV3's stated motivation**. They present presence gating as preventing hallucination of absent classes. Measured on LoveDA, its single largest effect is suppressing the background class. Their paper does not report this, and unlike the causal claim we abandoned, this one is directly measured.

Self-refutation #2: "road is a hub" was measuring road's perimeter

Our own published analysis had found that roads attract every class, and had built a weighting scheme on it. Then we checked the statistics rather than the conclusion.

The PMI formula compares an *observed* quantity against an *expected* one. We were observing **boundary frequency** but computing the expectation from **area**. These are different measures. A class shaped like thin ribbons has far more boundary per unit area than one in fat blocks — so the formula was rewarding high-perimeter classes for their geometry, with no semantics involved. Road is the thinnest class in LoveDA.

PAIR	AREA-BASED (PUBLISHED)	BOUNDARY-BASED (CORRECT)	VERDICT
road – barren	+2.32	+0.17	collapses
road – forest	+1.96	+0.06	collapses
barren – forest	+0.72	–1.11	flips sign
building – water	–4.66	–2.83	survives

The premise survived; the ranking did not. Mean signal recomputed correctly is **0.574 bits** against a **0.003 permutation floor** — 216×, still decisive. But "road is a hub" was an artefact, and the real hub is *agriculture*. We verified the diagnosis on synthetic data: a corpus where one class is a thin ribbon by construction gives it +3.00 under the wrong formula and +0.23 under the right one. This correction generalises beyond our project — anyone doing co-occurrence analysis on segmentation masks will hit it — and it became one of the paper's secondary contributions.

Two discard mechanisms, not one

A pixel can end up labelled background two ways: it fell below the threshold, or background genuinely won the vote. We separated them exactly.

MECHANISM	PIXELS	SHARE
(A) fell below the threshold	303,706,238	94.0%
(B) background won the vote outright	19,378,177	6.0%

Every single (B) pixel is **water**, concentrated in **24 tiles** — large open-water scenes. And (B) is **unreachable by any threshold whatsoever**, because lowering a threshold cannot change which class won a vote. Part of the residual lies outside what threshold tuning can address at all. That sharpened the motivation from "the obvious fix is expensive" to "and some of it the obvious fix cannot touch".

THE MOTIVATION SECTION, COMPLETE

Three blunt interventions, all measured, all failing: lower the threshold (–5.54, 1.73 wrong per right) · remove presence gating (–11.97, no net recovery) · do nothing (29.68% discarded). Every global knob that reaches the residual costs more than it returns, so any recovery has to be *selective* and *semantic*. Two refutations make a stronger motivation than one.

Act IV — The thesis dies

25-27 AUGUST · COMMITS `DC05235` → `286B643`

This act builds the thing the project was founded on, tests it properly, and finds it does not earn its place. It is the most important act, because everything good came after it.

The prior was built, and it is accurate

We mined the co-occurrence matrix from SAM 3's own confident predictions, using no labels — the honest, deployable version. The worry (written down back in Act I) was **circularity**: if the model is systematically unsure about *barren*, the mined matrix under-represents every barren relationship, and we then use that matrix to label the barren regions the model missed. The prior would work against the classes that need help most.

VALIDATION GATE	RESULT
Gate 1 — circularity	✓ Passes. Mined matrix agrees with the ground-truth matrix at $\rho = +0.757$. Error does <i>not</i> concentrate on the discarded classes ($\rho = -0.257$). The concern is retired with a number.
Gate 2 — usefulness	✗ Fails, and against ground truth too.

Gate 2 is the interesting failure. It asked: does the matrix assign *low* compatibility to exactly the pairs the baseline confuses? The baseline's top confusion is *forest predicted as agriculture*. But the **true** matrix says forest and agriculture are *attracted* (+0.32). A perfect matrix would *reinforce* the error.

The reason is simple and fatal: **adjacency and confusability are the same signal**. Things that touch each other look alike at the boundary. This is not a mining defect that better engineering could fix.

The ceiling test

We then asked the fairest possible question: given regions we *know* should be relabelled, how well does the prior label them, compared to the dumbest possible alternative?

LABELLING METHOD	PIXEL ACCURACY
Always guess the most common class	38.6%
Just copy the biggest neighbour (prior switched off)	48.4%
Best mix of neighbour vote and mined prior	48.6%
Best mix using the <i>perfect</i> ground-truth matrix	48.7%
Pure co-occurrence, no neighbour vote	20.1%

THE THESIS IS DEAD

A **perfect** co-occurrence matrix adds **0.3 points** over copying the largest neighbour. Not the mined one — the perfect one. There is no engineering effort that recovers this. It works only where four or more classes border a region, which is **10% of the residual**, diluting to +0.40 overall.

Verdict: it is an **ablation row**, not a thesis. And we do not revive it.

What actually mattered instead: the shape of the regions

While testing the prior, we tested the regions it operates on. This turned out to be worth sixty times more.

REGION METHOD	COUNT	LARGEST REGION	PURITY	BEST POSSIBLE ACCURACY
Connected components	41,952	1,048,576 px	0.728	72.8%
SLIC superpixels	506,064	60,740 px	0.928	92.8%

The largest connected component was *an entire image tile* — one region containing several different classes, so no single label could ever be right for it. Switching to SLIC lifted the ceiling by **20 points**, against 0.3 for a perfect prior. Pixels in unlabelable regions fell from 12.5% to 1.1%.

And the real blocker: we cannot tell which pixels to fix

Relabelling is only useful if you know *which* background pixels deserve it. We tested nine different signals for that, scored by AUC — 0.5 is a coin flip, 1.0 is perfect.

SIGNAL	LOVEDA AUC
Confidence of the winning class	0.582
Confidence of the runner-up class	0.541
Score before presence gating	0.559
Presence score	0.434
Region mean confidence	0.576
Novelty against class prototypes	0.514 / 0.528
Texture / gradient energy	0.622

Empirical chance floor for this setup: ~0.53.

All at chance. Three hypotheses died here: there is no hidden second threshold in the confidence scores; presence gating is not hiding the signal (the pre-gating score scores *worse*); and novelty detection fails for a reason consistent with its own premise — a residual class has no compact region of feature space to be far from.

A NEGATIVE CONTROL CAUGHT A CONFOUND WE WOULD HAVE PUBLISHED

Run on **random-colour images**, where no signal can possibly exist, the novelty detector scored **0.966**. Cause: a region's mean colour has noise that shrinks as the region grows, so a "distance" feature partly measures region *size*, which correlates with the answer. Fixed with size-stratified scoring and unit-tested both ways (pure size-leak 0.503; genuine signal 0.926). Without the control, an AUC near 1.0 would have gone into the paper as a breakthrough.

And a ground-truth leak in our own code

The first version of the recovery evaluation scoped its regions to pixels the ground truth says are real classes. That hands the method advance knowledge of where to look, and immunity from damaging true background — exactly the protection that threshold relaxation does not get. It produced a clean, quotable **+3.47** that was not a result. It is now kept deliberately as an explicitly labelled *upper bound*.

The second dataset, and the mechanism

One dataset is a desk reject. We added **OpenEarthMap**. Almost everything disagreed — and one variable explained all of it.

	LOVEDA	OPENEARTHMAP
"background" share of ground truth	36.1% (catch-all)	0.84% (rare, genuine)
Real pixels discarded	29.68%	3.78%
Catastrophic tiles ($\geq 99\%$ discard)	198	0
Presence–discard correlation	−0.750	+0.094
Best detection AUC	0.622	0.913
Pixels the model commits to	55.7%	95.9%

THE MECHANISM, IN ONE SENTENCE

A catch-all class gives the model a plausible answer everywhere, so a strong runner-up carries no information. Where the vocabulary genuinely covers the scene, a strong runner-up means a real class was suppressed — and the residual becomes detectable.

Two consequences we had to accept immediately: our own headline **29.68% is a LoveDA number, not a general one**, and the presence-collapse correlation is a property of LoveDA's annotation scheme rather than of SAM 3. Both are labelled as such throughout.

But recovery still failed — and the way it "worked" was a trap

On OpenEarthMap, recovery gave **+2.28 mIoU**. Genuinely tempting. We broke it down per class before believing it.

CLASS	BEFORE	AFTER	Δ
background	17.13	39.80	+22.67
building	75.32	71.57	−3.75
tree	63.91	63.26	−0.64
road	45.88	45.32	−0.56
water	66.57	67.60	+1.02

A HEADLINE GAIN THAT IS NOT A SEGMENTATION IMPROVEMENT

The eight real classes **lost 2.11 IoU in aggregate**. The catch-all alone gained 22.67 — **110% of the total**. The method made land-cover classification *worse*, and damaged the baseline's best class. Averaging over nine classes converted that into a positive headline.

We never quote +2.28 without this table beside it. It became one of the most useful things in the paper, because it is the table a sceptical reviewer would otherwise construct themselves.

At the end of Act IV the project had: a dead thesis, a settled engineering result about region shapes, a two-dataset mechanism, and no method.

Act V — The pivot

28 AUGUST · COMMITS `CA64D15` → `4E3A6C8`

An earlier note in the project said "threshold tuning is closed". Re-reading it, the evidence behind that sentence was a sweep of **one global threshold**, which found nothing (+0.04). **Per-class thresholds had never been tested**. That oversight turned out to be the whole paper.

Step 1 — is there anything there? (the oracle bound)

First we cheated on purpose, to find the ceiling: choose the best possible threshold for each class using the answer key.

RUNG	FREE PARAMETERS	LOVEDA MIOU
Published threshold (0.5 for everything)	0	47.37
Best single global threshold	1	47.41 (+0.04)
Best per-class thresholds	6	48.83 (+1.46)

And crucially, the chosen thresholds spanned **0.170 to 0.595**.

THE INSIGHT THE WHOLE PAPER RESTS ON

One global threshold is not merely *suboptimal* for different classes — it is wrong for them **in opposite directions**. That is why sweeping a single value finds nothing: raising it helps some classes exactly as much as it hurts others, and the average cancels.

water, which is right 89.5% of the time it fires but finds only half of what is there, wants a threshold of **0.170**. It gains **+6.70 IoU** alone.

This was also the only gain in the project so far where **land cover genuinely improves**: the six real classes gained +8.63 in aggregate, against background's +1.59. Not the catch-all artefact.

Step 2 — the deployable version

An oracle is not a method. The deployable version fits one threshold per class by coordinate ascent on a calibration subset, and evaluates on completely disjoint tiles.

WHY THIS IS NOT CHEATING

SegEarth-OV3 **already** tunes its single threshold per dataset using labels — 0.5 for LoveDA, 0.1 for OpenEarthMap. ConInfer tunes theirs too (0.8 / 0.3 / 0.1). We use the same protocol with six parameters instead of one. **No model weights are trained**. It is parity, not a supervision leak — and the baseline is always recomputed on the very same held-out tiles.

LOVEDA, 5-FOLD	PUBLISHED T	FITTED T	Δ
fold 1	47.91	49.28	+1.37
fold 2	45.41	47.31	+1.89
fold 3	48.44	49.32	+0.88
fold 4	46.11	46.91	+0.80
fold 5	48.68	49.66	+0.98
mean			+1.18 $\pm$ 0.45

Every fold positive. 81% of the oracle bound captured. Land cover +8.30 in aggregate; the catch-all moves -0.01 — so this result is structurally immune to the artefact that caught three earlier ones.

HOW MUCH LABELLING DOES IT COST?

CALIBRATION TILES	Δ	STD. DEV.	WORST DRAW
10	-2.14	1.99	-5.59
50	+0.08	0.97	-0.99
100	+0.54	0.71	-0.57
200	+0.79	0.35	+0.43
400	+1.21	0.16	+1.04

~200 tiles is where every draw turns positive. Below 50 it actively hurts. That is an honest, quotable deployment cost.

Step 3 — prove it on the real pipeline, not on our arithmetic

Everything above was computed from a cached histogram rather than by running the model, so it could all have been a bookkeeping illusion. We patched the actual segmentor to accept a threshold *vector*, fitted on 200 tiles, and ran the official evaluation script on the 1,469 disjoint tiles.

	PREDICTED FROM CACHE	MEASURED BY EVAL.PY
Published $\tau = 0.5$	47.16	47.16
Per-class τ	48.35	48.35
Δ	+1.18	+1.18

Both absolute values reproduce exactly; every per-class delta agrees to within 0.04, inside the cache's known rounding noise. **This validated the histogram as a measurement instrument**, so every threshold sweep and oracle bound in the project inherits the guarantee.

A BUG THE VERIFICATION TEST CAUGHT

The bin-edge calculation used truncation: $29/200 \times 200$ evaluates to 28.999999999999996, which truncates to 28. The function was scoring a threshold one bin below the one it reported. **No recorded number changed** — only 7 of 201 grid values were affected and none had ever been chosen — but a *deployed* threshold could have been wrong. That is precisely what this step exists to catch.

Step 4 — can we do it without labels at all?

The obvious next question. Eleven separate attempts, all failing, and the reason turned out to be more interesting than a twelfth attempt would have been.

Our initial explanation was: *the oracle exploits per-class precision, and precision requires labels by definition*. We tested that instead of asserting it, using eight label-free statistics.

LABEL-FREE PROXY	P VS PRECISION	P	Δ MIOU GAINED
mean_conf	+0.943	0.017	−0.18
gate_ratio	+0.943	0.017	−0.05
cross-head agreement	+0.886	0.033	+0.06
presence score	+0.657	0.175	+0.42
random control (200 draws)	—	—	p95 = +0.58
oracle bound	—	—	+1.24

OUR STATED REASON WAS WRONG, AND THE TRUTH IS STRONGER

Precision IS predictable without labels — four proxies rank the classes correctly, two of them at $p \leq 0.033$. **And it buys nothing.** The best row in the table is below a random control's 95th percentile.

Because precision *does not determine the threshold*: the correlation between a class's precision and its ideal threshold is -0.429 with $p = 0.419$, and the relationship is not even monotone. *water* has the highest precision and the lowest threshold; *road* has middling precision and the *highest* threshold.

The real bound: the right thresholds solve a *coupled* multi-class objective. Raising one class's threshold pushes pixels into background, which changes background's score, which changes every other class's optimum. **No per-class scalar can express that, measurable or not.** This is exactly why a six-parameter fit works and every one-parameter rule fails.

That reframes the negative from "we could not measure the quantity" to "we can measure it, and it is the wrong quantity" — which explains eleven failures instead of merely recording them, and generalises to any training-free pipeline that thresholds a per-class score.

Worth noting: the cross-head prediction was *written down before the run* — committed as "the one remaining candidate, and it will fail anyway, for the coupling reason". It turned out to be a good precision estimate and moved mIoU by $+0.06$. Exactly as predicted.

Act VI — From correlation to cause

31 AUGUST – 1 SEPTEMBER · COMMITS `BB61A36` → `DC78F71`

We now had a method and a mechanism. Both had a soft joint, and a reviewer would find both. This act hardens them.

Soft joint 1: the +1.18 is a rural result

LoveDA splits into urban and rural halves that differ 2× in discard rate. We ran the same five-fold protocol *within* each.

DOMAIN	TILES	Δ MIOU	FOLDS POSITIVE
rural	992	+2.77 ± 0.92	5 / 5
urban	677	+0.10 ± 0.39	2 / 5

Urban is not distinguishable from zero. We never quote +1.18 without this breakdown now. But the decomposition rescues the story rather than damaging it:

	RURAL	URBAN
real classes, aggregate	+18.61	+4.18
catch-all class	+0.79	−3.51
full mIoU	+2.77	+0.10

Land cover improves in both domains. Urban's flat headline is entirely the catch-all paying for the gain: $(4.18 - 3.51)/7 = +0.10$. This is the OpenEarthMap artefact *with the sign reversed* — there the catch-all inflated the metric, here it deflates it. Same mechanism, opposite direction, two datasets. That independent confirmation became a contribution in its own right (below).

We also found the thresholds are genuinely domain-specific — *road* wants 0.725 in rural and 0.225 in urban — and that **applying the wrong domain's thresholds is worse than no calibration at all** (both mismatched arms land below the published threshold). Pooling is not safe either: it fits one vector to two optima, which is *the same failure as the global threshold, one level up*. The method's own argument recurses on it. The practitioner rule: **calibrate on the distribution you will evaluate on**.

Soft joint 2: the mechanism was a correlation

Our strongest claim rested on comparing two datasets, and we had written in our own notes: "*this is a stratification, not a randomised intervention*". That sentence was the weakest joint in the strongest claim.

The fix came from noticing something about how the pipeline actually works.

THE OBSERVATION THAT MADE THIS FREE

In SegEarth-OV3, **every class is an independent forward pass with its own text prompt**. There is no softmax, no normalisation, no interaction between classes anywhere — the only cross-class operation in the whole pipeline is the final `argmax`.

Therefore **dropping a class from the vocabulary is exactly equivalent to dropping its channel from a saved array**. One cached run answers every vocabulary question. All thirteen experimental arms read *identical model outputs* and differ in nothing but the vocabulary — which is *better* than re-running, because re-running would have added sampling noise for nothing. Cost: **zero GPU hours**.

So we intervened on the causal variable directly: progressively merge real classes *into* the catch-all, raising its share from 0.84% to 58.20%, and re-measure detectability. Each dose has an **arity-matched control** that merges the same number of classes into *each other* instead — so class count changes identically and only the catch-all's share differs.

Twelve predictions were written into `PREREGISTRATION.md` and committed to git *before each run*, so the timestamps are independently verifiable.

SIGNAL	START (0.84%)	LARGEST DOSE (58.20%)	CONTROL	DOSE VS CONTROL
winning-class confidence	0.794	0.582	0.710	2.5×
runner-up confidence	0.913	0.590	0.855	5.6×

Both signals degrade causally with catch-all share, and neither degradation is explained by class count. The intervention drove OpenEarthMap from 0.794 into the 0.58–0.62 band where LoveDA actually sits — it *reproduced* the observational regime rather than merely correlating with it.

Nine of the twelve predictions passed. We report the three failures. The dose–response is non-monotone inside its range, one control moved slightly more than allowed, and the predicted *rate* missed at two of three doses. So: the direction transfers, the rate does not. All stated in the paper.

A NEAR MISS WORTH TELLING

The first verdict announced "CAUSAL" on the strength of an AUC dropping to **0.208**. But AUC is *symmetric*: 0.208 is a detector of strength **0.792 with its sign flipped**, not a destroyed signal. The score was rewritten as `max(AUC, 1–AUC)` everywhere — and on rescoreing, that arm's signal was *flat*, not destroyed.

The diagnosis also revealed a design flaw: the original merge made the catch-all a *union of well-detected prompts*, which is unnaturally competent. We kept it (it was pre-registered) and added a second family that drops the prompts and relabels the pixels — which is the real situation. That is the family the paper quotes.

The reverse experiment fails, and the failure is the finding

The mechanism predicted that *removing* the catch-all from LoveDA's vocabulary should raise detectability. Prediction P8, committed in advance. It moved from 0.592 to **0.540** — the wrong way.

ARM	VOCABULARY	GROUND-TRUTH SHARE	EFFECT
OpenEarthMap dose arms	changed	raised	0.213 – 0.323
LoveDA reverse arm	changed	unchanged	–0.052
LoveDA controls	changed	unchanged	+0.000 / –0.041

THE DISSOCIATION

Every arm that changes the vocabulary without changing the label space moves ≤ 0.052 . Every arm that raises the catch-all's share of ground truth moves 0.213–0.323. The causal locus is the LABEL SPACE, not the prompt list.

Practical consequence: **you cannot fix a catch-all-heavy benchmark by editing the prompt list.**

Removing the prompt does not remove the pixels that are genuinely none-of-the-above; it only makes them unnameable.

We also learned the effect **saturates by about 35% share**, which honestly narrows the claim — LoveDA starts at 35.7%, already inside the saturated band, so its dose arm had no room to act. We say that rather than presenting a single clean monotone story.

The metric recommendation, and OpenEarthMap turning positive

We now had the catch-all artefact measured in **both directions** on two datasets. That is enough to make a recommendation rather than a complaint.

THE LEVERAGE, WHICH IS NOT OBVIOUS

mIoU is an **unweighted mean over N classes**, so the catch-all owns exactly **1/N of the metric no matter how meaningful it is** — 14.3% on LoveDA for a class the annotation guide defines as "everything else", and **11.1% on OpenEarthMap for a class covering 0.84% of the pixels**. A 22.67-point move in that one class is 3.24 mIoU before anything real has changed.

Recommendation: benchmarks with a catch-all class should report mIoU over the real classes *alongside* the headline.

Recomputing every table both ways produced the best surprise of the project.

	LOVEDA	OPENEARTHMAP
Δ full mIoU (as reported for days)	+1.16	+0.30 — "flat"
Δ catch-all-excluded mIoU	+1.36	+1.75

A SECOND POSITIVE DATASET, RECOVERED FROM OUR OWN FILING ERROR

OpenEarthMap had been written up as flat and dismissed as "a benchmark artefact" **for four days**. Under the recommended metric it gains **+1.75 — more than LoveDA does**. Its headline decomposes exactly: +1.56 from the eight real classes, –1.26 from the catch-all alone. The catch-all was *cancelling a real gain*.

This closed the thinnest part of the method claim, and it cost hours of CPU time.

Two more negatives, both closed properly

Is there a label-free rule for *when* calibration is worth paying for? This would have turned a scope limitation into a deployment criterion — "measure your discard rate; above X%, buy the labels". We stratified both datasets by a label-free residual measure. LoveDA came out **U-shaped** (the most *reliable* gain is in the *lowest*-residual stratum), OpenEarthMap gave the **opposite sign**, and a random control moved nearly as far. No rule.

ONE BOUND EXPLAINS BOTH HALVES

You cannot pick the thresholds without labels (eleven attempts, with the coupling reason) *and* you cannot predict whether picking them will pay without labels either (two datasets, opposite signs). **The ~200-tile cost is irreducible: it buys the answer and the question at the same time.** That is a cleaner closing statement than the deployment criterion we went looking for.

What is the rural/urban gap actually made of? Arithmetic, since mIoU is an unweighted mean: *water* (48%) and *forest* (37%) carry **85%** of it. Across all twelve (domain, class) cells, the **precision–recall gap** ranks which classes move ($\rho = +0.713$, $p = 0.013$) — a measured confirmation of the method's stated mechanism — while **precision alone explains nothing** ($+0.168$, $p = 0.60$), the same lesson arriving independently.

But we report the failure first: *water*, the largest contributor, has effectively identical share (11.6% vs 11.8%) and precision–recall gap (+35.0 vs +34.4) in both domains, and an **8× different gain**. So roughly half the gap still has no mechanism, and the paper says so before quoting the correlation.

Act VII — The competitor, and the best result in the project

1 SEPTEMBER · COMMITS `D36AC53` → `2028F2A`

ConInfer is the nearest published competitor. We cited it in the paper *without a number*, which is the first thing a reviewer asks about. One table row closes it.

The rule that governed the whole exercise

NOTHING FOR CONINFER TOUCHES THE WORKING ENVIRONMENT. EVER.

Their install instructions read `conda create -n ConInfer` followed by `pip install -r requirements.txt` — **with no `conda activate` between them**. Run as written with our environment active, that command installs into *our* environment and destroys the only working version combination in the project.

We wrote a setup script that **refuses to run** while the main environment is active, snapshots its package list, and can diff it afterwards. And a matching package list is *necessary but not sufficient* — the real gate is behavioural: the baseline must still produce 47.38.

What reproducing published work actually required

This is worth recording plainly, because it is a reproducibility observation rather than a complaint, and it is evidence the run is real:

- A file their `eval.py` imports on line 4 is **absent from their repository**. Copied from the upstream project.
- **Three dependencies** declared by neither their requirements nor the packages importing them.
- Their segmentor **hardcodes the first author's filesystem paths**. We repointed them with a reversible patch that edits paths only.
- Their pinned torch version **cannot build on this machine**, for exactly the reasons documented in week one. We used the stack that works and reported the deviation.
- Their weights require a **licence-gated satellite-pretrained backbone** that returns HTTP 403.

A SHORTCUT WE DELIBERATELY REFUSED

A converted copy of those weights exists in a different format. Loading it would have required hand-writing a parameter remap — and a remap that is slightly wrong loads *successfully* with half the network still randomly initialised, producing a plausible number from a broken model with no way to detect it. We did not do it. That is precisely the failure the rule "reproduce their published figure first" exists to prevent.

The reproduction, reported honestly

	THEIR PUBLISHED	OUR REPRODUCTION	GAP	DECISION
LoveDA (full official split)	39.33	36.99	−2.34	✓ use it, report both
OpenEarthMap (77% of split)	41.95	29.90	−12.05	✗ drop it

We eliminated label encoding, prompt order, image format, a feature flag, and alphabetical subset bias as causes, then **stopped deliberately** rather than spend the timebox chasing an unverifiable difference. The

split that reproduces closely is the one we hold in full; the one that fails badly is the one we hold 77% of. That is not a coincidence, and it decided which comparison goes in the paper.

THE FRAMING THIS FORCES, AND WE INSISTED ON IT

Our numbers are 11.54 points above ConInfer's on LoveDA. **Most of that margin is the backbone** — SAM 3 at 1024² against CLIP ViT-B/16 at 448² — not our contribution. **Our contribution is the +1.16 over SegEarth-OV3**. Presenting the 11.54 as ours would be exactly the error we built a whole paper on catching. We should not make it after that.

Also reported alongside: they are **~10× faster** at inference (2 minutes vs 24). That belongs in the comparison too.

And then the experiment that was worth more than the comparison row

The comparison answers "how do you compare to ConInfer?". But there was a better question the paper could not answer: **is per-class thresholding just a SAM 3 quirk?**

ConInfer thresholds per-class scores with a single global value — precisely the design our method replaces. So we cached their scores in our format (a read-only wrapper, zero edits to their source), and pointed our existing fitting scripts at it, at *their* operating point of 0.8.

Both gates passed first: the instrumented run printed 36.99 exactly, and the scores landed inside the expected range.

	SAM 3 (SEGEARTH-OV3)	CLIP + GMM (CONINFER)
5-fold Δ full mIoU	+1.18 $\pm$ 0.45	+2.51 $\pm$ 0.34
worst fold	+0.80	+2.22
Δ catch-all-excluded	+1.36	+1.94
calibration tiles needed	~200	~25
classes that improve	5 of 7	7 of 7

THE TWO THINGS THIS BOUGHT

- 1. The claim is no longer about SAM 3.** Per-class thresholding is a property of *any pipeline that thresholds per-class scores* — demonstrated on two architectures, two methods and two operating points. The gain is *larger* on the pipeline it was not developed for.
- 2. ConInfer stops being a competitor.** Our method *composes* with it: their pipeline goes 36.99 $\rightarrow$ 39.52 with our calibration on top. We do not beat the nearest published work — we improve it. That is a far better answer to "how is this different from ConInfer?" than any margin would have been.

And the negative that made the claim precise

We ran the same test on ConInfer + OpenEarthMap. It came out **null** (-0.39 ± 1.28). But the *way* it failed was informative:

OPENEARTHMAP	OUR SAM 3 PIPELINE	CONINFER (CLIP)
Δ full mIoU	+0.30	-0.09
Δ excluding catch-all	+1.75	+1.27
catch-all IoU, before $\rightarrow$ after	17.13 $\rightarrow$ 5.83	17.00 $\rightarrow$ 6.06

Two completely unrelated models, and the fit drives the catch-all to almost the same value. That is the *dataset's* metric pathology, not a property of either pipeline. So the claim is scoped correctly:

THE PRECISE, FALSIFIABLE STATEMENT

The correction is **not backbone-specific** — **it is dataset-specific**, and the axis is the catch-all's calibration. Where the catch-all is competently predicted (LoveDA, ~ 45 IoU), the fit improves land cover *and* the headline, on both architectures. Where it is pathological (OpenEarthMap, ~ 17 IoU), the fit improves land cover on both and the headline on neither, because the catch-all it must sacrifice owns a ninth of the metric.

Act VIII — The third dataset falsifies us

3 SEPTEMBER · COMMITS `9090411` → `EABE959`

Two datasets is thin. We registered for ISPRS Potsdam, downloaded it via CLI (the browser route was quoting ten hours), and converted it.

Before running a single inference pass, we computed a prediction **from the ground truth masks alone** and committed it to git, so the timestamp proves it came first.

PRE-REGISTERED PREDICTION	MEASURED
Residual size in the bracket 3.78–10.88%, point estimate ~4.5%	4.68% ✓ HELD
Detection AUC <i>higher than</i> 0.622, point estimate ~0.885	0.578 ✗ FAILED

The baseline reproduced at **57.83 against a published 57.8** — the third reproduced baseline and the tightest of the three — so this is not a setup error.

The size prediction gives three points, monotone in catch-all share: 0.84% → 3.78%, 4.29% → 4.68%, 36.1% → 10.88%.

The detectability prediction is simply wrong. **Potsdam has a small catch-all (4.29%) and detects like LoveDA (36.1%), not like OpenEarthMap (0.84%).** The base rate sits between the two, so this is not a base-rate artefact either.

OUR OWN PRE-REGISTRATION NAMED THIS OUTCOME AS DISQUALIFYING

Written before the run, in these words: *"If a dataset near 5% background behaves like LoveDA, or one near 30% behaves like OpenEarthMap, the mechanism is wrong and the paper's central claim needs rewriting."*

So we rewrote it. The claim splits in two:

CLAIM	STATUS
catch-all share → size of the residual	✓ three datasets, monotone, one pre-registered and confirmed
catch-all share → detectability	✗ falsified across datasets

Importantly, this does *not* overturn the causal intervention of Act VI. Both can be true: share is **a** cause of detectability *within* a dataset, and is **not** what determines it *across* datasets. The paper says exactly that, and no more.

THE DECISION WE ARE MOST CONFIDENT WAS RIGHT

We did **not** go looking for a variable that fits three points. Any story invented now would be fitted to the data it must explain, and the entire value of the pre-registration is that it was not. A falsified prediction, reported as falsified, is worth more to a reviewer than a repaired one.

The method on a third dataset

	PUBLISHED T	FITTED	Δ
full mIoU	57.87	58.47	+0.60
catch-all-excluded	66.05	67.09	+1.04

Five-fold: $+0.59 \pm 0.50$, four of five folds positive, but the spread covers zero. Reported as **marginal on full mIoU and positive on land cover**, which is what both columns say.

And a second honest complication: *car* carries the gain (+3.70) while *tree* does not — despite tree having a precision–recall gap of +54.7, larger than the LoveDA *water* gap (+34.8) that drove the entire LoveDA result. So the precision–recall asymmetry did *not* predict which class would move here. That **weakens** our own correlation from Act VI rather than confirming it, and it is reported directly beside it.

One thing did get stronger with every dataset. The catch-all class depresses the published headline before any method is applied:

DATASET	CATCH-ALL IOU	REAL-CLASS MEAN	HEADLINE DEPRESSED BY
LoveDA	45.50	47.68	0.31
OpenEarthMap	17.13	47.54	3.38
Potsdam	16.96	66.05	8.18

The bug this dataset found

A SILENT FAILURE THAT PRODUCED A PLAUSIBLE NUMBER

Two lines of the measurement code quietly encoded "*the catch-all is the lowest class value*". True for LoveDA and OpenEarthMap. **False for Potsdam, where clutter is the highest.**

The result: every tile reported **0.00% discard** while the aggregate printed a plausible **5.55%** — computed over the wrong set of rows, with the catch-all counted as a real class losing 34.68% of its pixels *to itself*. The headline even fell inside our pre-registered bracket.

The tell was not the headline. It was "**mean 0.00%, median 0.00%, max 0.00%**" **sitting next to a non-zero total**. Fixed, and verified to be a bit-exact no-op on the two datasets where the old code was right, so no earlier number changed.

This was the **fourth** dataset-specific assumption to survive in code whose whole job is to read its class list from the data. A dataset whose catch-all is not first was always going to find them.

Where this research area stands

The lineage we inherited

Understanding why our contribution has a place needs about ten years of context, compressed.

YEAR	WORK	WHAT IT ADDED
2015	FCN, U-Net	Dense per-pixel prediction. U-Net is still the remote-sensing workhorse.
2020	ViT	Images as sequences of patch tokens — the unit everything downstream operates on.
2020	DETR	Object "queries" and set prediction. <i>Its first author is also first author on SAM 3</i> , so the instance head is the same design lineage.
2021–22	MaskFormer / Mask2Former	Segmentation as mask classification. SAM 3's decoder is built on this.
2021	CLIP	The whole open-vocabulary paradigm — text and images in one space.
2023–25	SAM → SAM 2 → SAM 3	Class-agnostic segmentation, then video, then segmentation + recognition unified with a text-conditioned presence head.

MAKING CLIP PRODUCE DENSE MASKS WITH NO TRAINING — THE DIRECT ANCESTORS

WORK	CONTRIBUTION
MaskCLIP (2022)	First to pull dense features out of CLIP. The origin of this whole field.
SCLIP (2023)	Correlative self-attention. <i>SegEarth-OV3's code is based on this repository.</i>
ClearCLIP (2024)	The "remove pooling" line of attack.
ProxyCLIP / CorrCLIP (2025)	Use a second foundation model (DINO, SAM) to supply structure CLIP lacks.
GroupViT (2022)	Weakly-supervised grouping — conceptually closest to a clustering step.

THE REMOTE-SENSING BRANCH — OUR ACTUAL NEIGHBOURHOOD

WORK	DATE	WHAT IT DOES — AND WHERE IT LEAVES ROOM
SegEarth-OV	CVPR 2025	CLIP features on aerial images are 1/16 resolution, which is fatal for small objects. They train a lightweight upsampler on <i>unlabelled</i> images and subtract a global bias. Source of the dataset protocol everyone now uses.
SegEarth-OV3	Dec 2025	Our baseline. Replaces the whole CLIP stack with SAM 3. Two ideas: fuse the instance and semantic heads (things vs stuff), and gate by the presence score. Evaluated on 20 datasets.
ConInfer	Mar 2026	Nearest competitor. First to explicitly add context at inference for this task: fits a statistical model over DINOv3 features and fuses it with the vision–language prior. +2.80 mIoU. Purely <i>visual</i> context, at <i>patch</i> level. Its own limitations section names region-level context as future work.
OVRISBenchV1	2026	A unified evaluation protocol across eight datasets. Standardisation is arriving — a sign of a maturing field.
SkySense-O, AerOSeg, RemoteSAM	2025	The "just pretrain on remote-sensing data" counter-argument, and other SAM-based unified pipelines.

What the state of the field means for us

Three readings, and they shaped every strategic decision after Act IV:

1. **The field has moved from "make CLIP dense" to "bolt a foundation segmenter on".** That move happened in December 2025, three months before we started. Competing on architecture against groups with cluster-scale compute, from one 70 W card, is the losing game.
2. **Inference-time context modelling was claimed in March 2026.** The original novelty of our problem statement had a competitor before we opened the file. Our differentiators — semantic rather than visual context, region rather than patch, corpus-level rather than per-scene — were real. Then we measured them and they were worth 0.3 points.
3. **What nobody is doing is the evaluation and calibration question.** Everyone in this lineage tunes a global confidence threshold per dataset and mentions it in a config file. Everyone reports mIoU including a catch-all class that owns $1/N$ of the metric. Nobody had asked whether that threshold is the right *shape*, or what the catch-all is doing to the numbers. **That empty ground is where we ended up** — not by design, but by measuring carefully enough that the gap became visible.

OUR COMPARATIVE ADVANTAGE, STATED PLAINLY

We cannot out-compute the competition. What we can do is out-*measure* them: reproduced baselines on three datasets, validation gates that must pass before any number is believed, negative controls, oracle bounds on every negative result, pre-registered predictions, and a mechanism that *explains* results rather than reporting them. Every decision from Act V onward played to that.

How we worked — the rules that produced this

Two dozen individual measurements are less useful to explain than the handful of habits that generated them, especially since most of the habits were adopted *after* something went wrong.

1. Measure before deciding

Every design choice that survived was settled by a cheap experiment rather than an argument: boundary-length adjacency, signed PMI, the hierarchical matrix, SLIC over connected components. The co-occurrence premise was tested with a 40-line script *before* any method was built on it — and the ceiling test that killed the method took an afternoon. Finding that out in week three instead of week eleven is the difference between a pivot and a disaster.

2. Validation gates are hard stops

Any instrumented run must still reproduce **47.37 mIoU and 29.68% discard exactly**. If either moves, the patch changed behaviour and the measurement is void. This caught nothing dramatic — which is the point. It is why every downstream number can be trusted, and it is what let us treat a cached histogram as a measurement instrument after proving it matched `eval.py` to 0.04 per class.

3. Write down what the result will mean, before you have it

Used three times. The presence-gating counterfactual had both readings written in advance — and the outcome matched neither cleanly, which made the gap between "the number I expected to be decisive" and "the number that actually decided it" impossible to paper over. The vocabulary intervention had twelve predictions committed to git. The Potsdam prediction was computed from labels alone and committed before any inference.

Pre-registration is nearly free and it is the single strongest signal of good faith a reviewer can be given. Almost nobody in computer vision does it.

4. Always run the control that shows what "nothing" looks like

A random-proxy control (95th percentile +0.58) is why we know the best label-free rule at +0.42 is noise. Random-colour images are why we caught a detector scoring 0.966 where no signal can exist. An arity-matched control is what made the vocabulary experiment causal rather than suggestive. On OpenEarthMap the random strata moved nearly as far as the real ones — without the control, that would have been written up as a pattern.

5. Cache everything the GPU produces

One 24-minute run wrote a per-tile cache. After that, every threshold sweep, oracle bound, cross-validation and vocabulary arm was a CPU pass measured in seconds. The thirteen-arm causal experiment cost **zero GPU hours** because of this. On a 70 W card it is the difference between running the experiments and not.

6. Report both metrics, always

Adopted after the catch-all artefact was measured in both directions. It converted a complaint into a recommendation other people can adopt, and it recovered a dataset we had filed as flat for four days.

7. Read the generated prose against its own table

THE RECURRING FAILURE MODE OF THIS CODEBASE, STATED FOR THE RECORD

Four separate incidents. A verdict that reasoned in prose about "higher" and "lower" and swapped both branches. A test that printed "414% of the change" because two contributions had opposite signs. A pass/fail gate that asked only "is the mean positive" and passed a result with three of five folds negative. A summary that quoted a LoveDA number on three other datasets' outputs.

In all four, the tables were correct. Only the generated text was wrong. The numbers were never at risk; the *conclusions printed beside them* were. Every verdict now gets read against its own table, and gates that can be phrased backwards get rewritten in a form that cannot be.

8. Push back on your own results hardest

Five findings were refuted by us before anyone else could:

FINDING	HOW IT DIED
"Road is a hub"	Wrong null model — it was measuring road's perimeter.
"Presence gating causes the collapse"	Counterfactual: recovery uncorrelated with presence (+0.018).
" +3.47 from selective recovery"	Our own code was scoping regions using the answer key.
"The oracle needs precision, which needs labels"	Precision is label-free. It is simply the wrong quantity.
"Catch-all share governs detectability"	Potsdam, against a prediction we had committed in advance.

Where the project stands today

The results, in one place

DATASET	BASELINE (OURS / PUBLISHED)	Δ FULL MIOU	Δ EXCL. CATCH-ALL
LoveDA — SAM 3	47.38 / 47.4	+1.18 $\pm$ 0.45	+1.36
LoveDA — ConInfer (CLIP)	36.99 / 39.33	+2.51 $\pm$ 0.34	+1.94
OpenEarthMap — SAM 3	44.19 / 42.9	+0.30	+1.75
Potsdam — SAM 3	57.83 / 57.8	+0.60	+1.04
LoveDA rural / urban split	—	+2.77 / +0.10	—

Land cover improves on every dataset and both architectures. Full mIoU moves only where the catch-all is competently predicted — which is itself one of the paper's findings, not a caveat hiding one.

Assets

- **Paper draft:** 7,671 words, 26 references, one placeholder left (the author block). Every number in it comes from a macro file so a figure cannot go stale in one section and current in another.
- **Supplementary material** and an Overleaf bundling script.
- **Seven figures**, each rendered by a script that prints every plotted number against its source table on render — a figure that has drifted from its data is worse than no figure.
- **Three reproduced baselines**, all within 0.03 of published where the split is complete.
- **Pre-registration files** with git timestamps.
- **A daily logbook** and $\sim 60,000$ words of measured results across five documents.

Known limits, stated because they will be asked

- The thresholds **do not transfer across a domain shift**. The wrong domain's thresholds are worse than no calibration. Calibrate on the distribution you will evaluate on.
- The LoveDA gain is **carried by the rural half**. Urban land cover improves but full mIoU does not.
- **Half of the rural/urban gap has no mechanism**. It is 85% *water* and *forest*, but *water* looks identical in both domains and gains $8\times$ more in one.
- The ConInfer reproduction is **2.34 points below their published figure**, and we report both numbers rather than the one that suits us.
- **Half the mechanism claim is falsified** by our own third dataset, reported as falsified.

What is left

TASK	COST	STATUS
Verify five reference entries against their PDFs	~1 hour	manual
Read the paper end to end in the real template	~2 hours	manual
Fill the author block	minutes	manual
Settle the title	—	deferred by choice
Optional: Vaihingen as a fourth dataset	~30 min GPU	would tell us whether Potsdam is the anomaly
Optional: the one label-free family not yet bounded	1 CPU day	publishable either way

Timeline

WHEN	WHAT
Sep – Dec 2026	Optional experiments, writing, figures
1 January 2027	Content freeze — no new experiments after this date
Jan – Feb 2027	Write, cut to length, internal review
~March 2027	Submit to CVPR EarthVision 2027 <i>(date not yet officially posted; verify the call for papers)</i>

Fallback venues, in order: **IGARSS 2027** (~January deadline), **IEEE GRSL** (rolling), and **IEEE TGRS** — which is rolling, so there is no deadline to miss, and is the natural home for a measurement paper with a causal experiment. With three datasets and the ConInfer comparison done, TGRS became plausible where it previously was not.

THE CONTENT FREEZE EXISTS FOR A REASON

Six months is enough time to do three good things and far too much time to keep finding a fourth. A project that misses a March deadline because it was still measuring in February has made an avoidable mistake, and the roadmap has warned about it since week four.

Closing — what the project actually is

It started as a plan to rescue thrown-away pixels using the spatial grammar of land cover. That plan is dead, and it was killed by our own measurement rather than by a reviewer: a *perfect* co-occurrence matrix is worth 0.3 points over copying the biggest neighbour.

What survived is smaller, sharper and better supported. **A single confidence threshold is the wrong shape for a multi-class open-vocabulary pipeline.** It is wrong for different classes in *opposite directions*, which is why nobody found it by sweeping one value. Fitting one threshold per class, on the same labelling budget the baseline already spends, and training no weights at all, recovers a real gain — on two architectures, two methods, and three datasets, and it *composes with* the nearest published competitor instead of beating it.

Around it sits a measurement study with a causal experiment, a pre-registered prediction that held, a pre-registered prediction that failed and is reported as failed, and a metric recommendation that other people can adopt.

THE ONE-LINE VERSION

We set out to fix what the model discards, and ended up fixing the rule that discards it.

The most valuable thing in the repository is not any single number. It is that every negative result carries an oracle bound saying how much was actually available, so a reader learns where the headroom is rather than only that we failed to reach it — and that five of our own findings were refuted by us, in writing, with the numbers that killed them.

Appendix — reference list

OUR BASELINE AND COMPETITORS

- **SegEarth-OV3** — *Exploring SAM 3 for Open-Vocabulary Semantic Segmentation in Remote Sensing Images*. arXiv:2512.08730. **The named baseline. Reproduced at 47.38 vs 47.4.**
- **SegEarth-OV** — CVPR 2025. Source of the dataset protocol.
- **ConInfer** — *Context-Aware Inference for Training-Free Open-Vocabulary Remote Sensing Segmentation*. arXiv:2603.29271. **Nearest competitor. Run, measured, and improved by our method.**
- **OVRISBenchV1** — *Towards Realistic Open-Vocabulary RS Segmentation*. arXiv:2604.15652.

ARCHITECTURAL LINEAGE

- FCN (Long et al., 2015) · U-Net (Ronneberger et al., 2015)
- ViT (Dosovitskiy et al., 2020) · DETR (Carion et al., 2020)
- MaskFormer / Mask2Former (Cheng et al., 2021, 2022)
- CLIP (Radford et al., 2021) · DINOv2 (Oquab et al., 2023)
- SAM (Kirillov et al., 2023) · SAM 2 (Ravi et al., 2024) · **SAM 3 (Carion et al., 2025)**

TRAINING-FREE OPEN-VOCABULARY SEGMENTATION

- MaskCLIP (Zhou et al., 2022) · SCLIP (Wang et al., 2023) · ClearCLIP (Lan et al., 2024)
- ProxyCLIP (Lan et al., 2025) · CorrCLIP (Zhang et al., 2025) · GroupViT (Xu et al., 2022)

CONTEXT MODELLING — THE CLASSICAL ANSWERS WE POSITION AGAINST

- SLIC superpixels (Achanta et al., 2012) · Felzenszwalb–Huttenlocher (2004)
- DeepLab with CRF post-processing (Chen et al., 2017) · CRF-as-RNN (Zheng et al., 2015)
- EncNet — Context Encoding for Semantic Segmentation (Zhang et al., 2018)

REMOTE-SENSING SPECIFIC

- SkySense-O (Zhu et al., 2025) · AerOSeg (Dutta et al., 2025) · RemoteSAM / InstructSAM (2025)

DATASETS USED

- **LoveDA** — 7 classes, urban/rural split, 1,669 validation tiles at 1024², 0.3 m resolution. Catch-all 36.1%.
- **OpenEarthMap** — 9 classes, global coverage, 384 of 500 validation tiles. Catch-all 0.84%.
- **ISPRS Potsdam** — 6 classes, 2,016 tiles at 512², 5 cm resolution. Catch-all 4.29%.
- **iSAID** — *evaluated and rejected*. Its catch-all is 97.11% of ground truth *and* maximally confusable, so it would confirm the ordering and discriminate nothing. Documented rather than downloaded.